MIT WORLD PEACE UNIVERSITY, PUNE

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

CINEMATCH: AN AI-POWERED MOVIE DISCOVERY, HYBRID RECOMMENDER SYSTEM & OTT STREAMING PLATFORM

A Formal Academic Project Report Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Technology in Artificial Intelligence & Data Science

Candidate Name:	**Pauravee Phuke**
Program & Degree:	B.Tech in Artificial Intelligence & Data Science
Class & Division:	Division 9, Batch I2
Course / Subject:	Python Programming & Applied Machine Learning
Academic Year:	2026 – 2027
Live URL:	**https://pauraveephuke2-a11y.github.io/**

ABSTRACT

In the contemporary landscape of digital entertainment, content discoverability has emerged as a central challenge in human-computer interaction and applied artificial intelligence. While streaming providers host catalogs spanning tens of thousands of media titles, viewers are consistently overwhelmed by the paradox of choice. This academic project presents CineMatch, a full-stack, production-grade movie recommendation and Over-The-Top (OTT) streaming platform developed using modern data engineering and web technologies.

The core intelligence of CineMatch leverages Natural Language Processing (NLP) content-based filtering powered by Term Frequency-Inverse Document Frequency (TF-IDF) vectorization and Cosine Similarity metrics, reinforced with user-item collaborative filtering heuristics. The application features two interchangeable, high-performance architectural presentations: (1) a reactive Python Streamlit application with rich analytics, and (2) a zero-dependency, ultra-fast client-side single-page application built on HTML5, CSS3, and modern ECMAScript (ES6+). The system further incorporates an industry-standard OTT dark-mode UI, responsive cinema video playback, public web stream aggregation, dynamic Unified Payments Interface (UPI) QR code generation linking to merchant settlement ID 7499525420@axl (Axis Bank), automated VIP membership activation, and a continuous deployment pipeline hosted on GitHub Pages.

1. INTRODUCTION & PROBLEM FORMULATION

1.1 Background and Motivation

Commercial video-on-demand services generate massive data matrices of titles, metadata, and viewership signals. Traditional keyword searches and broad categorical carousels fail to capture subtle narrative nuances, director aesthetic styles, or momentary emotional desires. Machine learning-driven recommender systems bridge this semantic gap by modeling multidimensional feature spaces and projecting prospective user satisfaction.

1.2 Problem Statement

Designing an effective recommender system requires solving several classical machine learning challenges: (1) high dimensionality and semantic sparsity of unstructured textual descriptions, (2) the cold-start problem where new or niche films lack substantial viewer ratings, and (3) frontend latency constraints where recommendations must be inferred in real-time on resource-constrained mobile client browsers without high-cost cloud computing dependencies.

1.3 Core Project Objectives

- **NLP Feature Engineering:** Automated feature synthesis pipeline using Pandas and Scikit-Learn that synthesizes plot, cast, genres, and mood into rich synthetic documents.
- **Mathematical Modeling:** Formulate rigorous TF-IDF vectorization and cosine similarity distance matrices bounded strictly between [0.0, 1.0].
- **Dual Full-Stack Implementations:** Construct both a Python Streamlit reactive application and a standalone zero-dependency HTML5/CSS3/JavaScript client-side web application.
- **Industry-Standard OTT Features:** Incorporate hero banners, mood quizzes, dynamic search, watchlist management, and embedded video streaming players.
- **UPI Payment Integration:** Build a functioning UPI-only subscription gateway with dynamic QR code rendering targeting Axis Bank merchant ID 7499525420@axl.
- **Automated Verification & CI/CD:** Establish dual GitHub repository synchronization, GitHub Pages continuous deployment, and Playwright headless browser end-to-end verification.

2. THEORETICAL FOUNDATIONS & MATHEMATICAL FORMULATIONS

2.1 Term Frequency - Inverse Document Frequency (TF-IDF)

TF-IDF is an established information retrieval statistic that reflects the relative importance of a token t within a document d relative to an overall corpus D. In CineMatch, each film's synopsis, director, top cast, and mood tags are concatenated into document d:

• **Term Frequency:** TF(t, d) = f(t, d) / ∑_{t' ∈ d} f(t', d)
• **Inverse Document Frequency:** IDF(t, D) = ln((1 + |D|) / (1 + |{d ∈ D : t ∈ d}|)) + 1
• **Composite TF-IDF Weight:** W(t, d, D) = TF(t, d) × IDF(t, D)

2.2 Vector Space Model & Cosine Similarity Metric

Each film is mapped to a vector in high-dimensional feature space. The semantic proximity between film A and film B is computed as the cosine of the angle between their vector representations:

cos(θ) = (A · B) / (||A||■ × ||B||■) = ∑_{i=1}^n (A_i × B_i) / (√∑ A_i² × √∑ B_i²)

Because token frequencies are strictly non-negative (A_i ≥ 0), the cosine similarity score is bounded in [0.0, 1.0], where 1.0 signifies exact thematic concordance and 0.0 represents complete orthogonality.

3. COMPREHENSIVE INVENTORY OF TECHNOLOGIES & CODEBASE ARTIFACTS

index.html	HTML5, CSS3, ES6+ JS	Production OTT streaming platform, JS TF-IDF engine, UPI QR generator, cinema video player modal.
app.py	Python 3.14 (Streamlit)	Analytical reactive web app with roulette randomizer, 30-sec quiz, and watchlist.
recommender.py	Python (Scikit-Learn)	NLP TF-IDF vectorizer and 30x30 global pairwise cosine similarity computation.
collaborative_filter.py	Python (Pandas)	Memory-based user-item collaborative filtering engine simulating peer user ratings.
movies.csv	Data (CSV)	30 curated blockbusters across 12 features (IMDb ratings, votes, runtime, mood, cast, synopsis, poster).
verify_live.py	Python (Playwright)	Automated headless browser test executing full end-to-end verification of the live deployed web portal.
launch_public.py	Python / Cloudflared	Spawns Streamlit backend and creates zero-password global HTTPS Cloudflare tunnel for edge access.

4. FRONTEND ENGINEERING & OTT STREAMING CLIENT

The production web frontend in index.html is engineered as a zero-dependency Single Page Application (SPA). It avoids external CDN links to ensure 100% uptime across all cellular networks (Jio, Airtel). Key technical capabilities include:

• **Cinema Video Modal:** 16:9 responsive player container encapsulating YouTube HD streams and CDN fallbacks.
• **Client-Side Vector Engine:** Instantaneous tokenization and cosine similarity calculation executing in < 3ms.
• **UPI Payment Engine:** Canvas QR generation protocol targeting Axis Bank settlement ID **7499525420@axl** with deep linking.
• **VIP Subscription Lifecycle:** Unlocks premium streaming status, renders gold VIP active badge, and spawns celebratory particle confetti.

5. CI/CD DEPLOYMENT & AUTOMATED QUALITY ASSURANCE

The application is continuously deployed on GitHub Pages across two independent endpoints:
1. **Root User Site:** https://pauraveephuke2-a11y.github.io/
2. **Project Mirror:** https://pauraveephuke2-a11y.github.io/cinematch/
Automated end-to-end testing via Playwright Chromium verified HTTP 200 response codes, desktop/mobile responsive rendering, modal lifecycles, and QR code rendering without errors.

6. CONCLUSION & REFERENCES

CineMatch illustrates a complete, modern data science and full-stack software engineering project, bridging theoretical NLP foundations with an intuitive, mobile-responsive consumer experience at MIT-WPU Pune.

- Ricci, F., et al. (2015). Recommender Systems Handbook. Springer.
- Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825-2830.
- NPCI (2024). Unified Payments Interface Linking Specifications & Protocol Architecture.